VAISHNAVI PUJALA

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SUMMARY

Computer Science professional with a genuine passion for Software Engineering and full-stack development. Combines hands-on engineering experience across web, backend, and distributed systems with published research in applied AI, bringing both the technical depth to design scalable solutions and the analytical mindset to solve complex engineering problems

EDUCATION

Georgia State University, Atlanta, GA| M.S. Computer Science, GPA: 3.3/4.0 — Aug 2024 – May 2026

Coursework: Machine Learning, Neural Networks, Deep Learning, LLMs, Federated Learning

G. Narayanamma Institute of Technology and Science, India | B.Tech Computer Science, GPA: 8.43/10 — Aug 2020 – Jul 2024

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, Go, C, C++, SQL, HTML, CSS
Frameworks & Libraries: React.js, Next.js, Node.js, WebSockets, FastAPI, Flask, Django, Flutter, Tailwind CSS, PyTorch, TensorFlow, Keras
Cloud & DevOps: AWS(S3, SQS, SNS, Lambda), GCP, Docker, Kubernetes, GitHub Actions, CI/CD, Linux, Bash
Databases: PostgreSQL, MongoDB, DynamoDB, Redis, NoSQL
Tools & Concepts: REST APIs, Microservices, System Design, OOP, Unit Testing, Agile, Kafka, NLP, LLMs, Tableau, Power BI, Data Structures

EXPERIENCE

Graduate Research Assistant| Georgia State University, USA — Aug 2025 – Present

- Pioneered a privacy-preserving federated learning pipeline for decentralized datasets, architecting Domain Adversarial Neural Networks (DANN) with a Gradient Reversal Layer (GRL) to strengthen cross-domain generalization across federated clients while ensuring sensitive data remains on local devices
- Benchmarked and evaluated federated optimization algorithms including FedAvg and FedProx, conducting rigorous experiments to analyze convergence behavior, communication efficiency, and model performance across federated clients to assess the scalability and robustness of decentralized training

Graduate Teaching and Lab Assistant| Georgia State University, USA — Aug 2024 – May 2025

- Instructed and mentored 100+ undergraduate students across CSC3320 System-Level Programming and Python-based data analytics courses, delivering hands-on coding exercises, practical lab sessions, and in-class demonstrations covering C, C++, memory management, pointers, data manipulation, and visualization to strengthen applied programming skills and overall academic performance
- Collaborated with faculty to craft lab materials, regression and classification exercises, and grading workflows across both courses, analyzing student performance data to refine assessment strategies, sharpen feedback mechanisms, and streamline evaluation processes for improved learning outcomes

Computer Vision & ML Engineer| Atal Incubation Centre, AIC-GNITS Foundation, India — Jul 2023 – Jul 2024

- Constructed and optimized a computer vision pipeline for copy-move image forgery detection using CNN and VGG-16, executing image preprocessing, feature extraction, and model training with Python, TensorFlow/Keras, and OpenCV to achieve 90% detection accuracy across evaluation datasets
- Delivered a full-stack ML inference system by integrating a Flask-based backend with an interactive HTML/CSS/JavaScript frontend, enabling real-time image upload and forgery localization — featured in Springer ICCCE 2024

AI/ML Research Associate| Innovation and Incubation Cell - GNITS, India — Aug 2022 – Jun 2023

- Developed a heart disease prediction system comparing Random Forest, Naïve Bayes, and Logistic Regression, applying feature selection, hyperparameter tuning, cross-validation, and data preprocessing to achieve 91% accuracy with Random Forest — recognized in Springer ICDSMLA 2023
- Performed exploratory data analysis and produced interactive visualizations using Plotly, Seaborn, and Matplotlib to surface health risk patterns and support integration of predictive models into AI-driven healthcare analytics workflows

PROJECTS

URL Shortener | Python, FastAPI, PostgreSQL, Redis, React, TypeScript, Docker — Dec 2025 – Feb 2026

- Engineered a full-stack URL shortener with JWT auth, Redis rate limiting, click analytics (device, geo, referrer), and a React and TypeScript dashboard using FastAPI, PostgreSQL, and Docker, handling 500+ concurrent requests with sub-100ms response time

Distributed Task Queue System | Python, FastAPI, Redis, PostgreSQL, React.js, WebSockets, Docker — Sep 2025 – Nov 2025

- Architected a Celery-like task queue with Redis-backed priority queues, 3 async workers, exponential backoff retries, dead letter queue, and a real-time React dashboard over WebSockets, containerized with Docker, processing 1,000+ tasks per minute under load

Netflix Content Intelligence Dashboard | Python, Pandas, Scikit-learn, NLP, Streamlit — Jun 2025 – Aug 2025

- Spearheaded an end-to-end data analysis pipeline across 8,807 Netflix titles using TF-IDF vectorization, KMeans clustering, and a cosine similarity recommendation engine, deployed as a live Streamlit web app with sub-second query response time

PUBLICATIONS

- **Springer Journal:** An effective Prediction of Heart Diseases using Machine Learning Techniques, ICDSMLA 2023 – Machine Learning
- **Springer Journal:** An approach to Detect Copy Move Forgery Detection using Deep Learning Techniques **,** ICCCE 2024 – Computer Vision